Excitations and defects in quantum materials

Learning outcomes

- Non-equilibrium response can be computed with the equilibrium ground state
- Responses allow to predict instabilities and see emergent quasiparticles
- Impurities in quantum materials allow to see the response of the ground state

Today’s plan

- The response in classical magnets
- Linear response for many electrons and electronic instabilities
- Impurity effects in quantum matter

Response in materials



In general, for a certain observable *A* we will have

$$A(x, x', t, t')$$

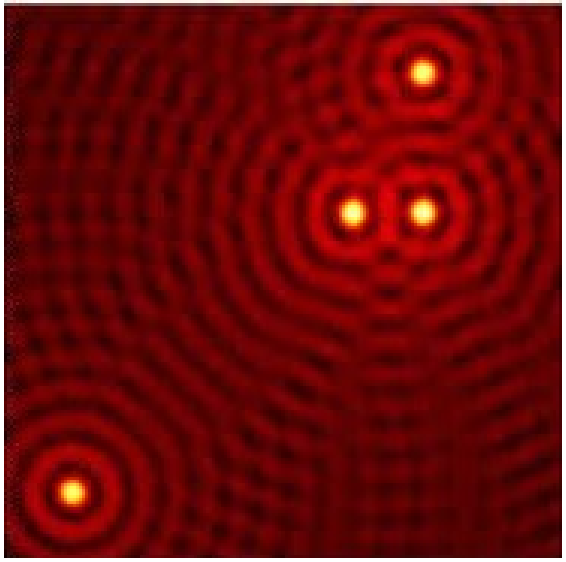
Which kind of perturbation?

→ Charge (dielectric and conductivity)
→ Spin fluctuation (magnetic susceptibility)
→ Heat (thermal conductance)

Types of responses

Impurities in a metal

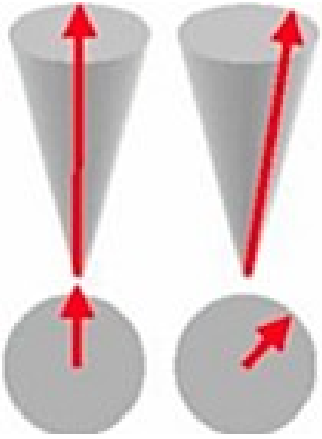
Friedel oscillation

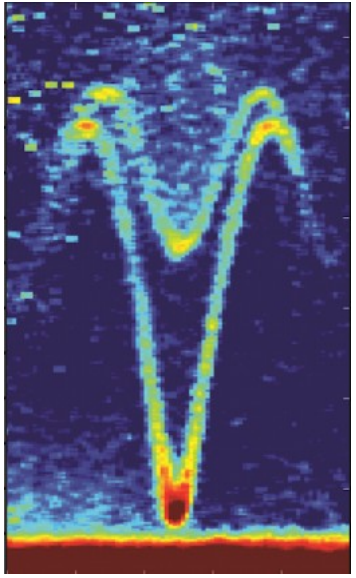


Information on the Fermi surface

Neutron scattering

(Inelastic neutron-matter process)





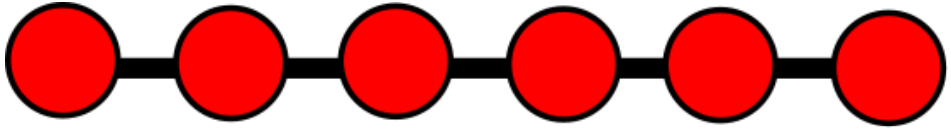
Information on magnetic excitations

Response in periodic systems

We will deal with systems with translational symmetry

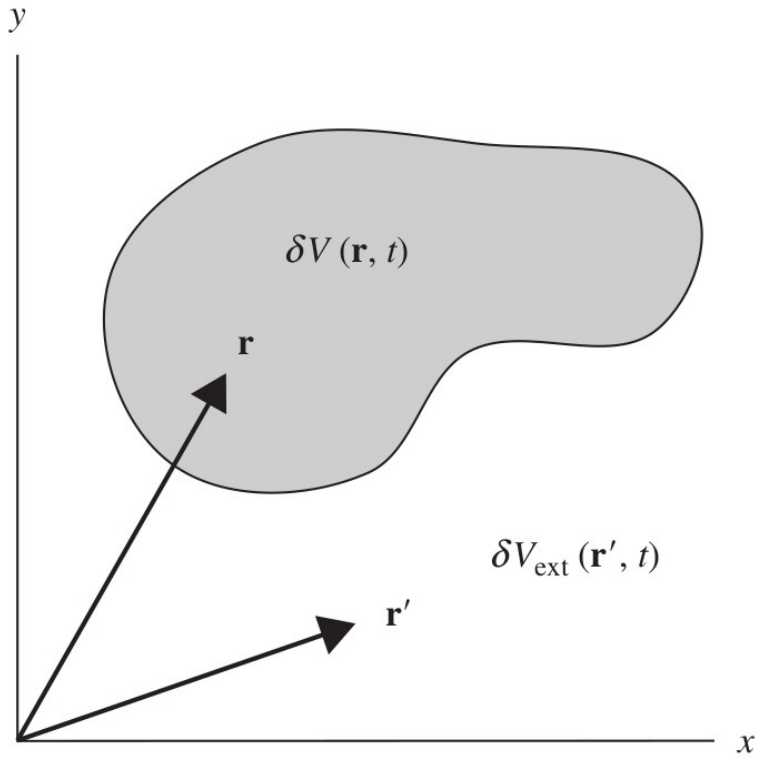
$$A(x, x', t, t') = A(x - x', t - t')$$

(like the one dimensional infinite periodic chain)



$$H = \sum_{i=-\infty}^{\infty} c_i^{\dagger} c_{i+1} + h.c.$$

A simple example: charge response



$$\delta V(\mathbf{r}, t) = \int \epsilon^{-1}(\mathbf{r}, \mathbf{r}', t - t')\delta V_{\text{ext}}(\mathbf{r}', t')d\mathbf{r}'dt'$$

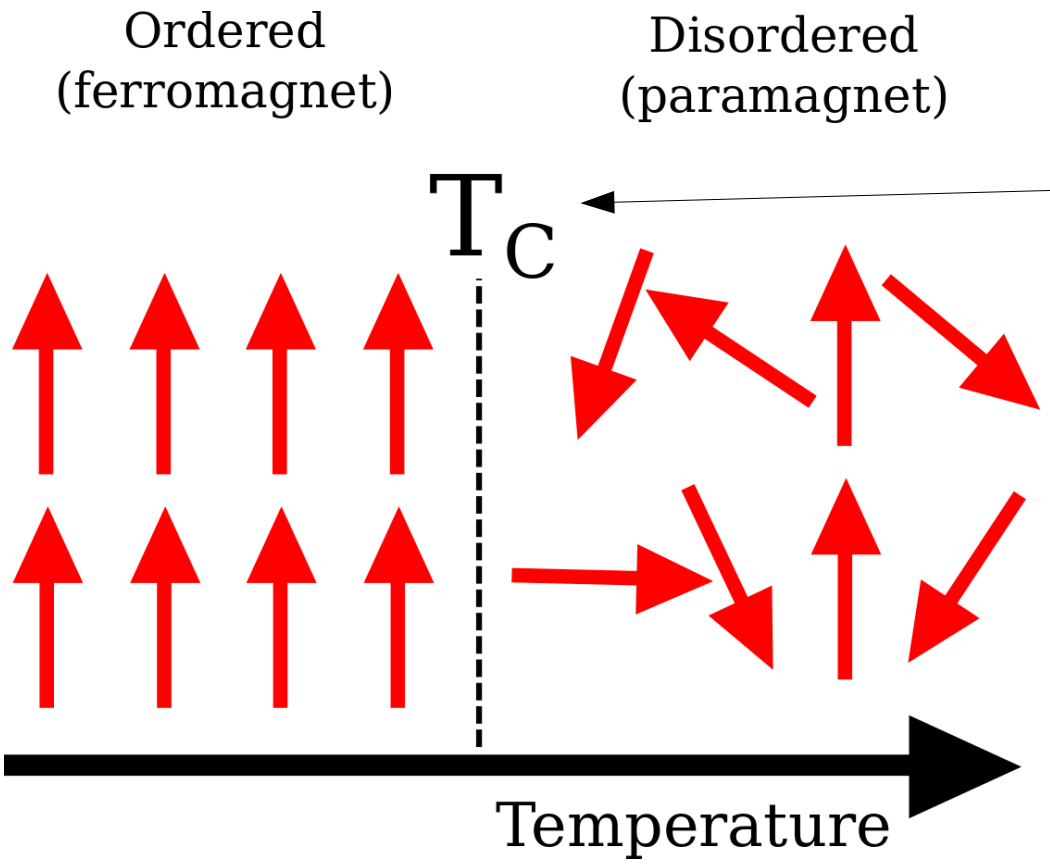
Change in potential

Linear response

Perturbation

Magnetic response in a classical magnet

Classical phase transitions



Critical temperature

Can we know if a system orders from its high temperature behavior?

Magnetic susceptibility



$\vec{M}$

$\vec{B}$

How does the magnetization of a material change when we apply an external magnetic field?

$$\vec{M}(\vec{B})$$

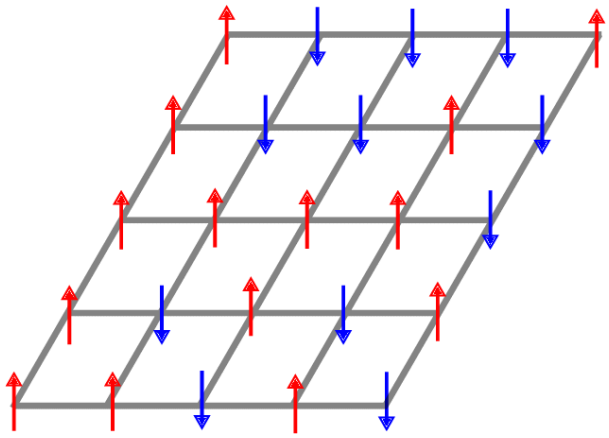
at high temperatures?

For small fields, we can perform a linear expansion

$$M_z(B_z) = \chi B_z$$

$$\chi = \frac{\partial M_z}{\partial B_z}$$

Magnetic susceptibility



Take an Ising model

$$H = J \sum_{\langle ij \rangle} S_i^z S_j^z + \sum_i B S_i^z$$

The high-temperature susceptibility (mean field) is $\chi = \frac{\partial M_z}{\partial B_z}$

Paramagnet
for $J = 0$

$$\chi \sim \frac{1}{T}$$

Ferromagnet
for $J < 0$

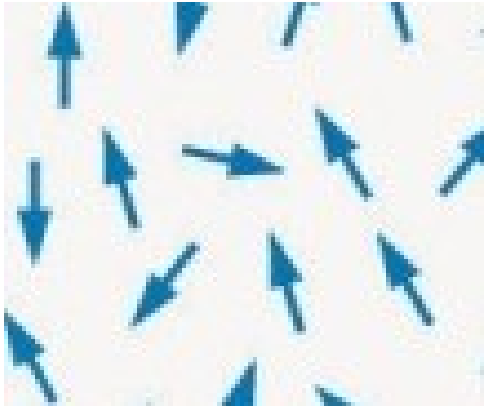
$$\chi \sim \frac{1}{T - T_C}$$

Antiferromagnet
for $J > 0$

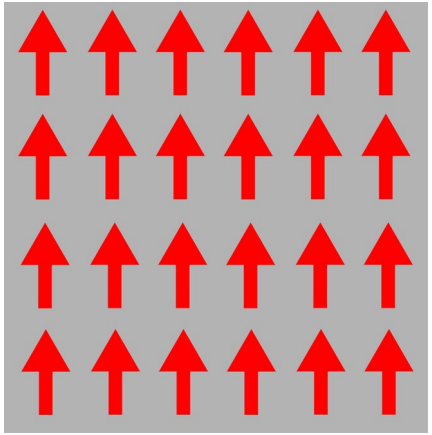
$$\chi \sim \frac{1}{T + T_C}$$

Three types of magnetic systems

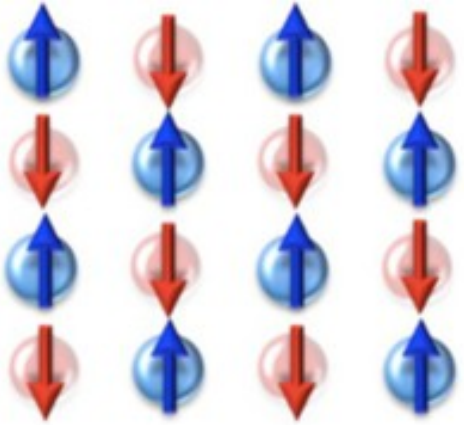
Paramagnet
(any T)



Ferromagnet
(low T)



Antiferromagnet
(low T)



Paramagnet

$$\chi \sim \frac{1}{T}$$

Ferromagnet

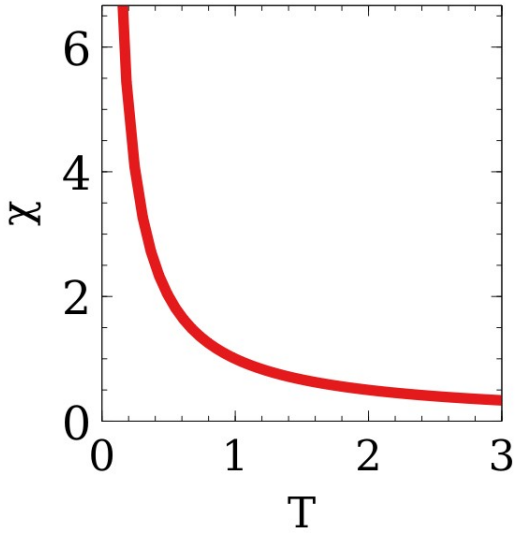
$$\chi \sim \frac{1}{T - T_C}$$

Antiferromagnet

$$\chi \sim \frac{1}{T + T_C}$$

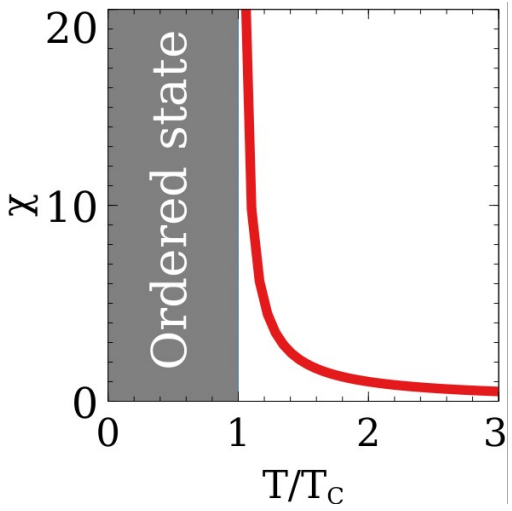
The high-temperature susceptibility

The high-T response tells us how the system behaves at low temperatures



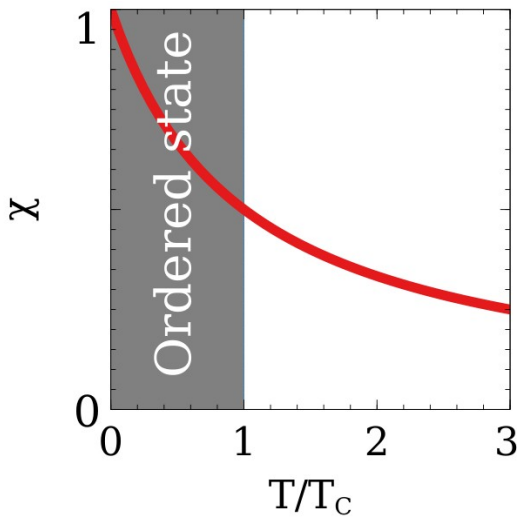
Paramagnet

$$\chi \sim \frac{1}{T}$$



Ferromagnet

$$\chi \sim \frac{1}{T - T_C}$$



Antiferromagnet

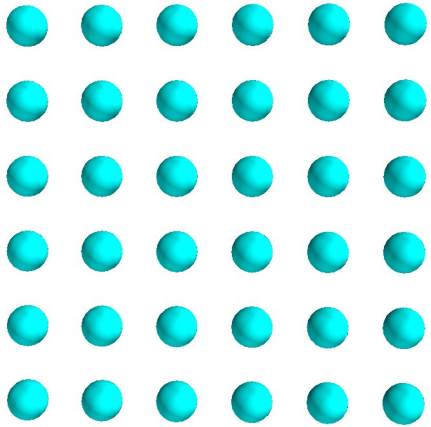
$$\chi \sim \frac{1}{T + T_C}$$

Linear response in a quantum system

Unperturbed and perturbed Hamiltonian

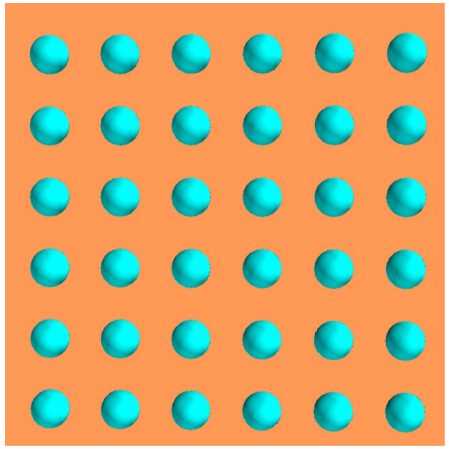
Unperturbed Hamiltonian

$$H_0$$



Perturbed Hamiltonian

$$H = H_0 + \lambda V$$



Can we infer the properties of a perturbed Hamiltonian from the unperturbed one?

Perturbation theory

Take a Hamiltonian that includes a perturbation V

$$H(\lambda) = H_0 + \lambda V$$

Every “well behaved” function is linear for small argument

Energy	***Observables***	***Wavefunctions***
$E_{GS}(\lambda)$	$\langle A \rangle(\lambda)$	$\lvert \Psi_k \rangle(\lambda)$

We will use a Taylor expansion in terms of the parameter λ

Observables in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Full Hamiltonian

Unperturbed Hamiltonian

Perturbation

For any observable

$$A(\lambda) \approx A(\lambda = 0) + \lambda\chi$$

Full value

Unperturbed value

Contribution from the perturbation

Response of the system

$$\frac{\partial A}{\partial \lambda} \equiv \chi$$

Eigenfunctions in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Eigenfunctions after the perturbation

$$|\Psi_i\rangle(\lambda) = |\Psi_i\rangle(\lambda = 0) + \lambda \sum_{j \neq i} \alpha_j |\Psi_j\rangle(\lambda = 0)$$

Eigenfunction after the perturbation

Eigenfunctions before the perturbation

Eigenfunctions in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Correction to the eigenfunctions to first order

$$|\Psi_i\rangle(\lambda) = |\Psi_i^0\rangle + \lambda \sum_{j \neq i} \alpha_j |\Psi_j^0\rangle$$

Coefficients

$$\alpha_j = \frac{\langle \Psi_i^0 | V | \Psi_j^0 \rangle}{\epsilon_i^0 - \epsilon_j^0}$$

Notation

$$|\Psi_i^0\rangle \equiv |\Psi_i\rangle(\lambda = 0)$$

$$H_0 |\Psi_i^0\rangle = \epsilon_i^0 |\Psi_i^0\rangle$$

The perturbed eigenfunctions can be expressed in terms of the unperturbed ones

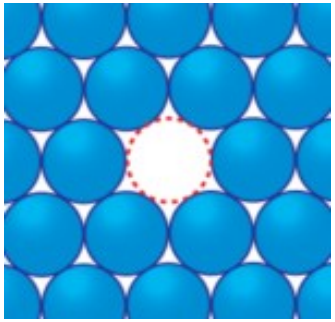
Breakdown of perturbation theory

$$H(\lambda) = H_0 + \lambda V$$

Is any full many-body ground state perturbative in terms of λ ?

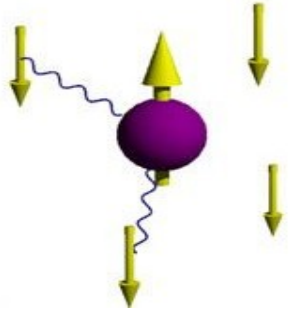
Certain quantum states are intrinsically non-perturbative

Impurity in a metal (Orthogonality catastrophe)



The many-body ground state Is orthogonal to the original

Quantum spin in a metal (Kondo effect)



The Kondo temperature is non-perturbative

Superconductivity (BCS limit)

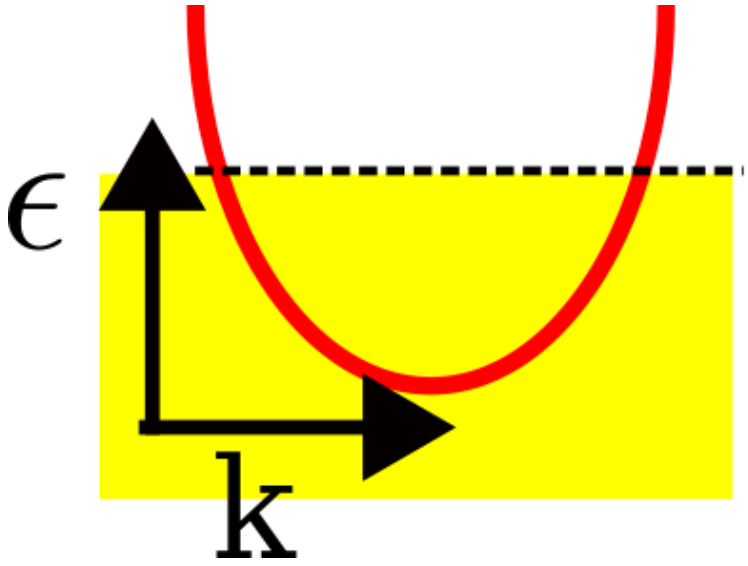


The superconducting gap is non-perturbative

Many-body response

The Fermi sea

Take a certain band-structure



$$\epsilon_F = 0$$

$$H = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k$$

States below the Fermi energy are filled

The Fermi surface is the set of k-points that cross the Fermi energy

$$\{\vec{k}\} \text{ with } \epsilon_{\vec{k}} = \epsilon_F$$

Ground state of an electronic system

Lets take a fermionic Hamiltonian in diagonal form

$$H = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k \qquad \{\Psi_i^\dagger, \Psi_j\} = \delta_{ij}$$

The many body eigenstate with minimum energy (ground state) is

$$|GS\rangle = \prod_{k \in occ} \Psi_k^\dagger |\Omega\rangle$$

Where *occ* is the set of *k* with $\epsilon_k < 0$ with $\Psi_k|\Omega\rangle = 0$

$$H|GS\rangle = E_{GS}|GS\rangle$$

The total ground state energy is $E_{GS} = \sum_{k \in occ} \epsilon_k$

Expectation values for the many electron system

Ground state

$$|GS\rangle = \prod_{k \in occ} \Psi_k^\dagger |\Omega\rangle$$

Hamiltonian

$$H = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k$$

By definition

$$\langle GS|\Psi_\alpha^\dagger \Psi_\alpha|GS\rangle = 1 \quad \text{if } \alpha \in occ$$
$$\langle GS|\Psi_\alpha^\dagger \Psi_\alpha|GS\rangle = 0 \quad \text{if } \alpha \notin occ$$

Ground state energy

$$E_{GS} = \langle GS|H|GS\rangle = \sum_{k \in occ} \epsilon_k$$

Perturbation theory for the many electron system

$$H = H_0 + \lambda V$$

$$|GS\rangle = \prod_{k \in occ} \Psi_k^\dagger |\Omega\rangle$$

Local charge

$$V = c_0^\dagger c_0$$

$$H_0 = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k$$

The change in charge in the system is defined as

$$\langle c_n^\dagger c_n \rangle - \langle c_n^\dagger c_n \rangle_{GS} \equiv \chi(n)\lambda$$

Given that we have translational symmetry, we will work with the Fourier transform

$$\chi(q) = \sum_n e^{iqn} \chi(n)$$

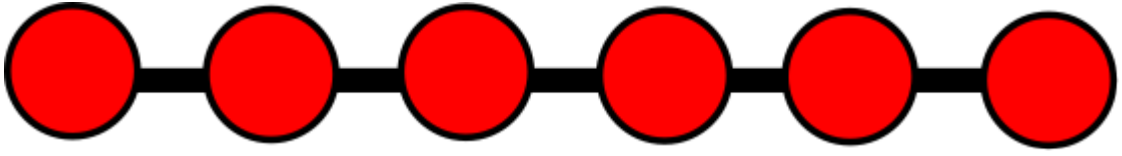
The Lindhard formula

Using the formulas from perturbation theory we can show

$$\chi(q) = \sum_k \frac{n_k - n_{k+q}}{\epsilon_k - \epsilon_{k+q}}$$

$$n_k = \langle GS | \Psi_k^\dagger \Psi_k | GS \rangle$$

$$n_k = 0, 1$$



$$H = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k$$

Instabilities and nesting

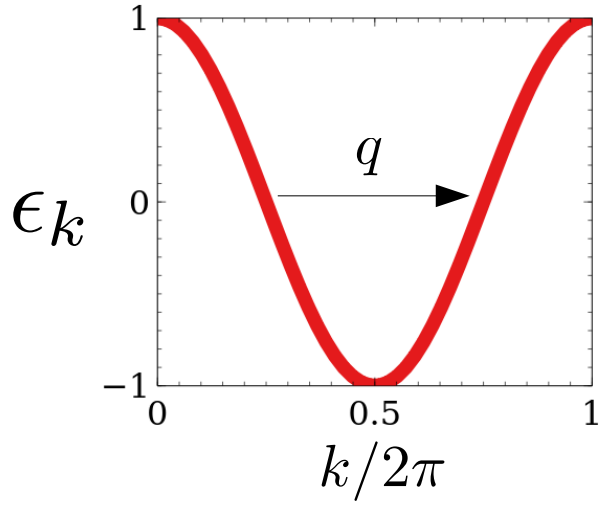
Take the form of the static response

$$\chi(q) = \sum_k \frac{n_k - n_{k+q}}{\epsilon_k - \epsilon_{k+q}}$$

The response diverges for

$$\epsilon_k = \epsilon_{k+q}$$

1D band structure



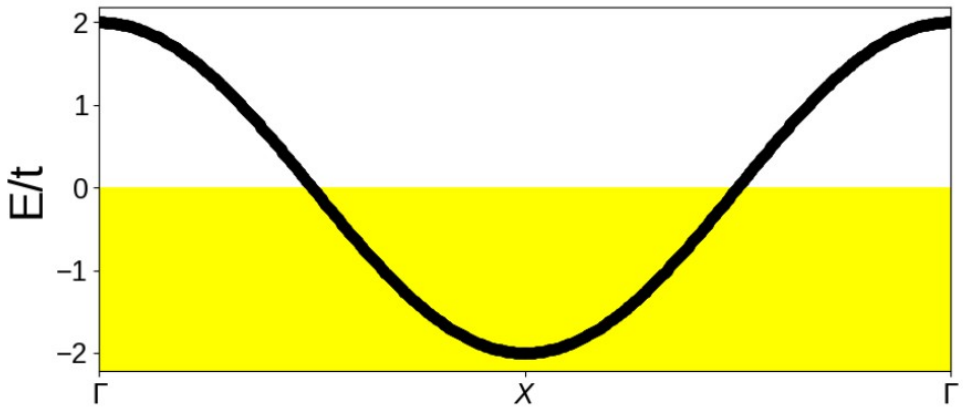
A very small perturbation creates a huge response in the system, yielding an instability

Instabilities and nesting

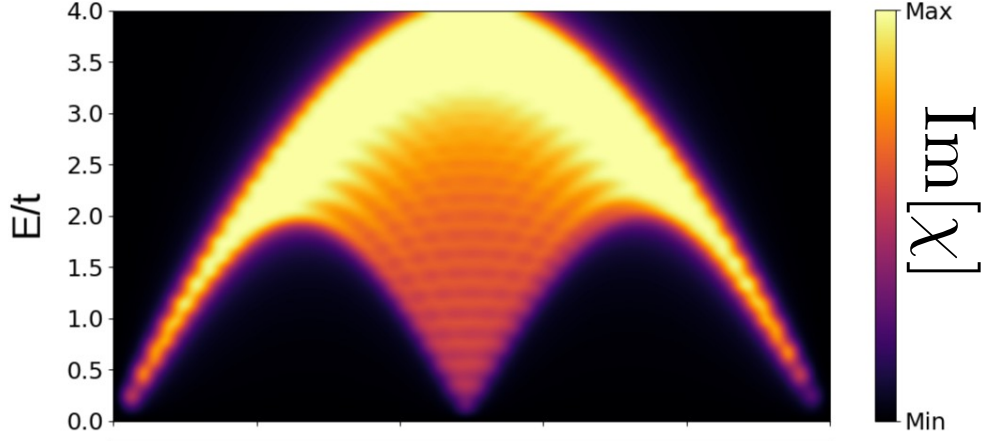
The frequency and momentum dependent charge response function

$$\mathrm{Im}[\chi(q,\omega)] = \mathrm{Im}\sum_k \frac{n_k - n_{k+q}}{\epsilon_k - \epsilon_{k+q} - \omega - i0^+}$$

Electronic structure



Charge response

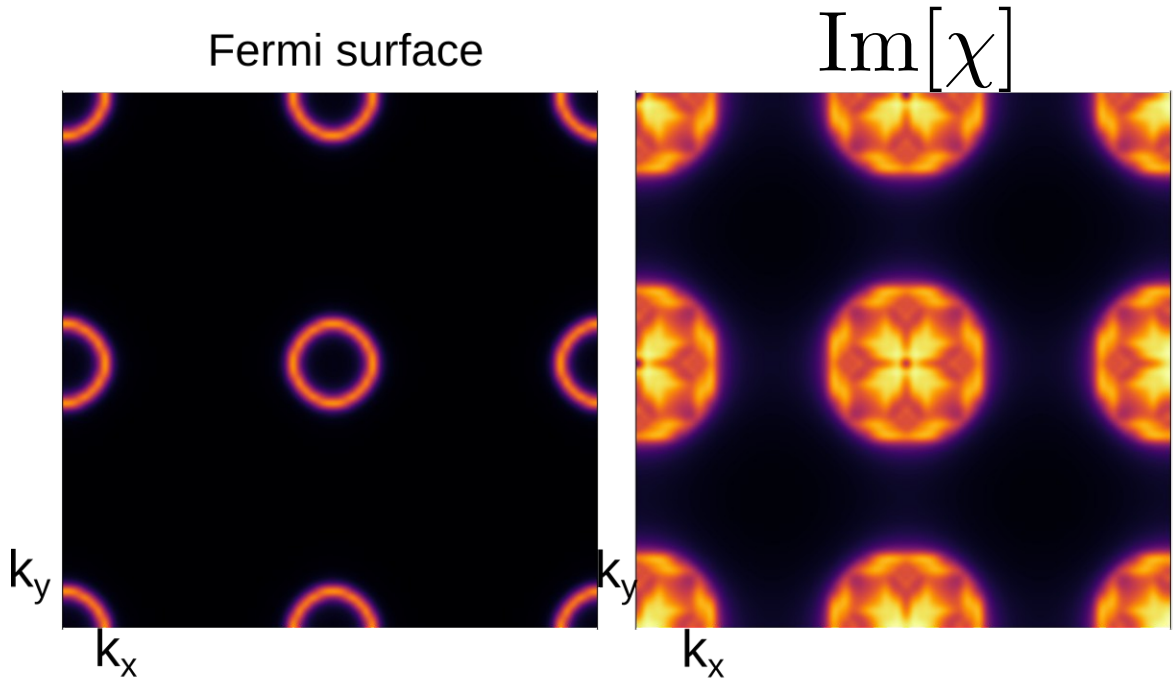


The charge response depends directly on the electronic structure

Dielectric response in the square lattice

Taking a square lattice

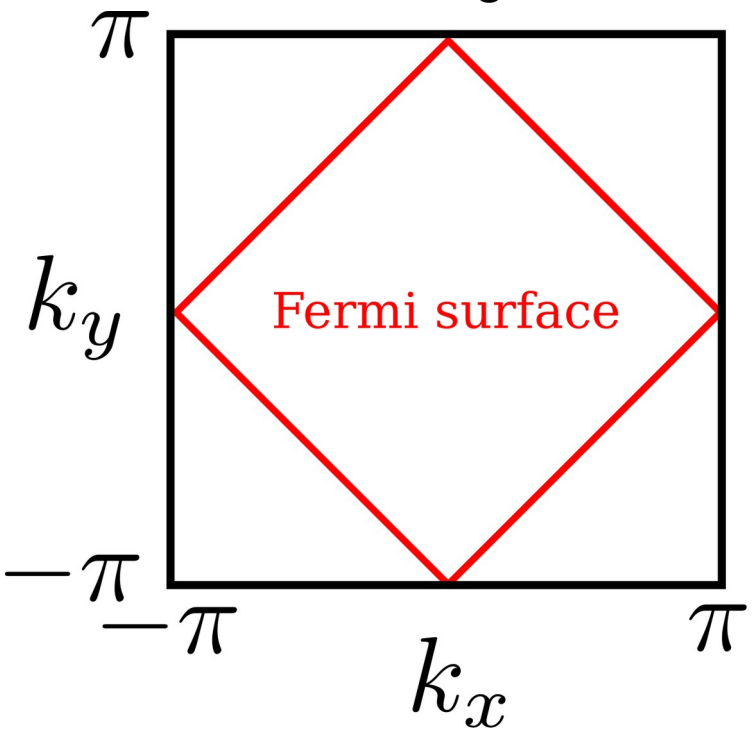
$$\epsilon(\mathbf{k}) = 2\cos k_x + 2\cos k_y$$



The response function directly reflects the Fermi surface

Instabilities in the square lattice

The Fermi surface of the square lattice at half filling has this form



Fermi surface for $\epsilon_k = 0$

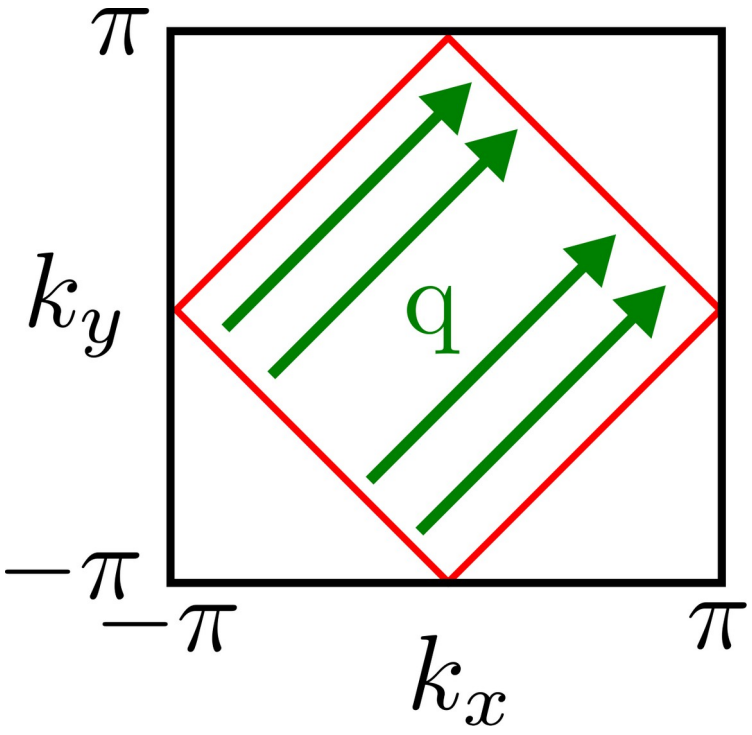
$$\chi(q) = \sum_k \frac{n_k - n_{k+q}}{\epsilon_k - \epsilon_{k+q}}$$

$$n_k = 0, 1$$

At which q-vector does the response diverge?

Instabilities in the square lattice

There is a common q vector that links same energy states



Charge/spin response

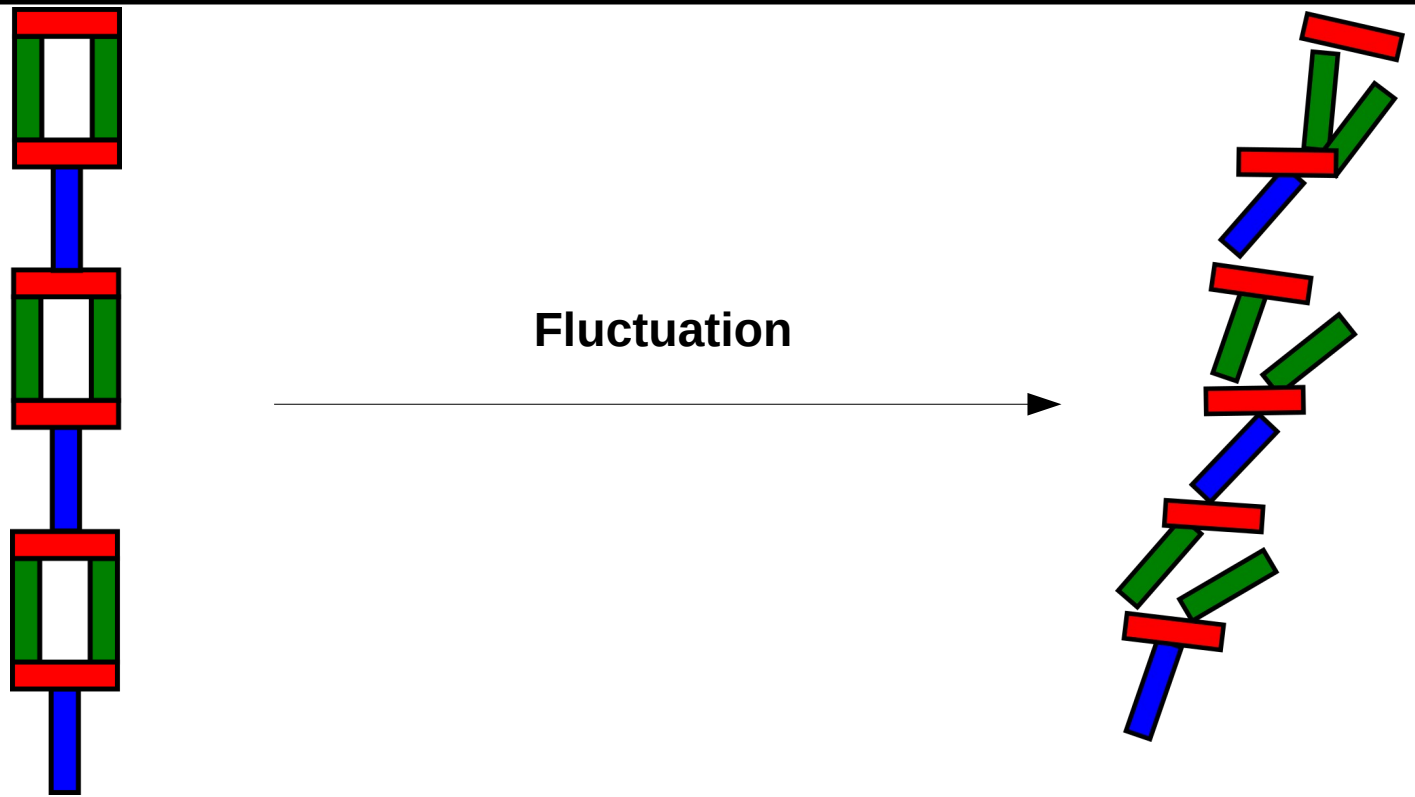
$$\chi(q) = \sum_k \frac{n_k - n_{k+q}}{\epsilon_k - \epsilon_{k+q}}$$

$$n_k = 0, 1$$

The response diverges at

$$q = \pi(1, \pm 1)$$

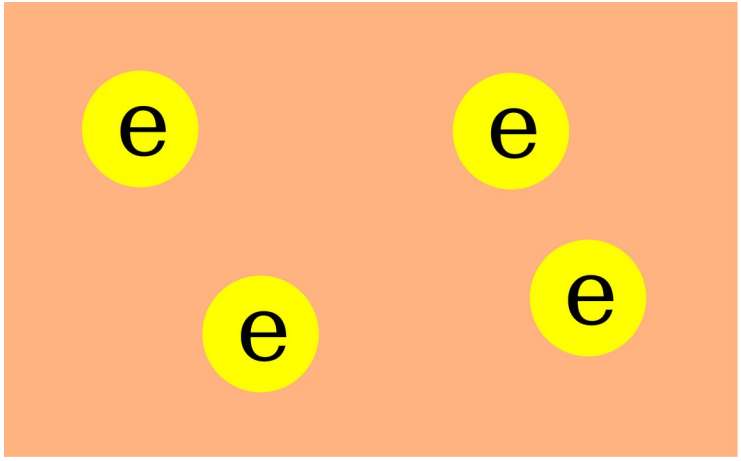
Fluctuations and instabilities



What is the condition for a small perturbation to “break down” a ground state?

Instability in an electronic system

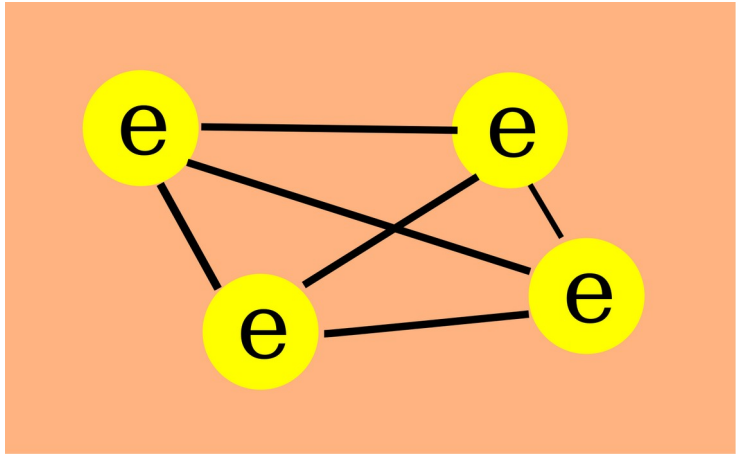
Without interactions



Response of the non-interacting system

χ_0

With interactions



Response of the interacting system

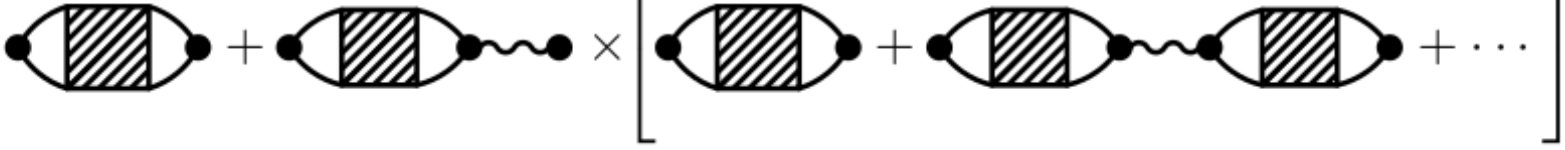
χ

How to know when a ground state breaks down due to interactions?

Perturbation theory in the interaction

$\chi(q) =$ 

$=$ 

$=$ 

$=$  $= \chi_0(q) + \chi_0(q)v(q)\chi(q)$

Instabilities, nesting and interactions

Response of a system in the presence of interactions

Free Hamiltonian ***Interactions***

$$H = \sum_k \epsilon_k \Psi_k^\dagger \Psi_k + \sum_{k,k'q} v_q \Psi_{k+q}^\dagger \Psi_k \Psi_{k'-q}^\dagger \Psi_{k'}$$

The random phase approximation (RPA) for the charge response

Response of the full Hamiltonian → $\chi(q) \approx \dfrac{\chi_0(q)}{1 - v(q)\chi_0(q)}$ ← Response of the free Hamiltonian

Plasmon quasiparticles

Taking the interacting dynamical response function

$$\chi(q,\omega) \approx \frac{\chi_0(q,\omega)}{1 - v(q)\chi_0(q,\omega)}$$

Quasiparticles arise from the imaginary part of the response $\chi(q,\omega) \sim \frac{1}{z(\omega,q) + i\eta}$

Two channels for quasiparticles

$$\mathrm{Im}[\chi_0(q,\omega)] \sim \delta(\epsilon_k - \epsilon_{k+q})$$

Electron-hole excitations

$$\mathrm{Im}[\chi(q,\omega)] \sim \delta(\omega - (\alpha[v] + \beta[v]q^2]))$$

Plasma excitations

Magnetic instabilities of an electron gas

Spinful Hamiltonian with local interactions

$$H = \sum_{k,s} \epsilon_k \Psi^\dagger_{k,s} \Psi_{k,s} + \sum_k U c^\dagger_{n\uparrow} c_{n\uparrow} c^\dagger_{n\downarrow} c_{n\downarrow}$$

RPA for the spin response $\chi^s(0) \approx \frac{\chi^s_0(0)}{1 - U\chi^s_0(0)}$

Density of states (DOS)

$$D(\omega) = \sum_k \delta(\omega - \epsilon_k)$$

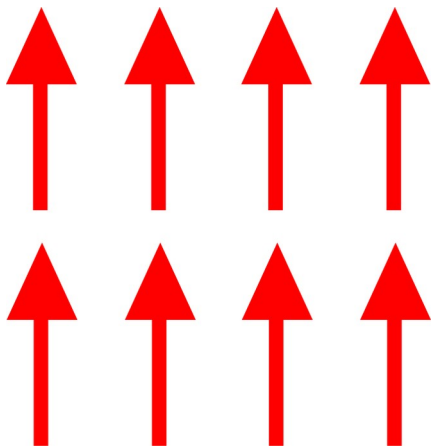
The Stoner criterion

$$D(\omega = 0)U = 1$$

A system can become magnetic when DOS times interactions is sufficiently large

Interacting instabilities in materials

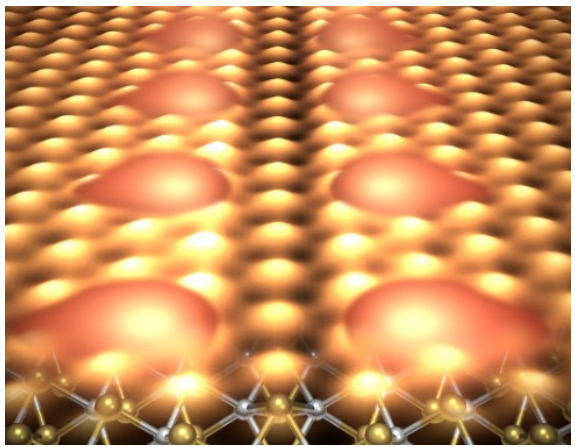
Magnetism



Instability in the spin response

(repulsive interaction)

Dimerization & charge density wave



Instability in the charge/phonon response

(repulsive interaction)

Superconductivity



Instability in the e-e scattering

(attractive interaction)

Impurities in quantum materials

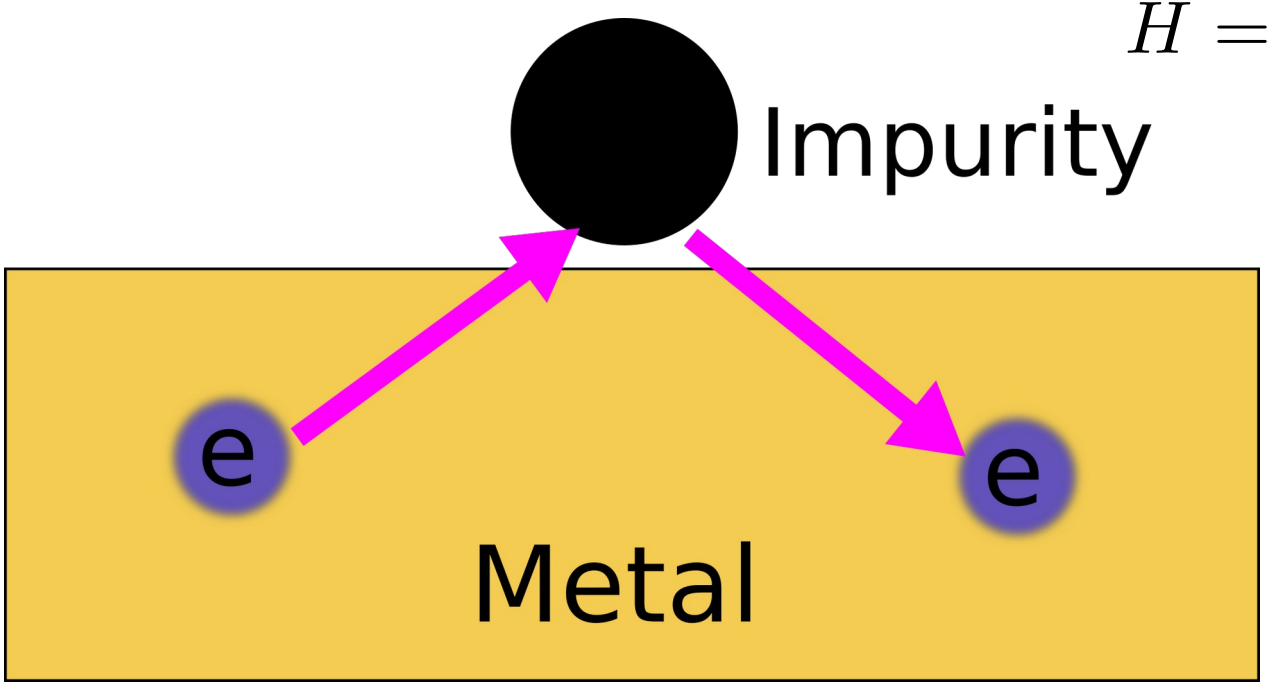
Impurities in quantum materials

- Adding an impurity in a quantum material allows to see the response of the ground state to a perturbation
  - Impurities in metals
  - Impurities in topological matter
  - Impurities in superconductors
  - Impurities in quantum magnets

Impurities in metals

$$H = H_{metal} + V c_0^\dagger c_0$$

perturbation



An impurity in a metal creates a perturbation mixing electronic states

Impurities in metals

Change in the charge distribution due to an impurity at the Fermi energy

$$\delta D(\epsilon_F, x) \sim \sum_{k,q \in FS} \langle 0|k\rangle|\langle q|x\rangle\langle 0|q\rangle|\langle k|x\rangle \sim \cos(2k_F x)$$

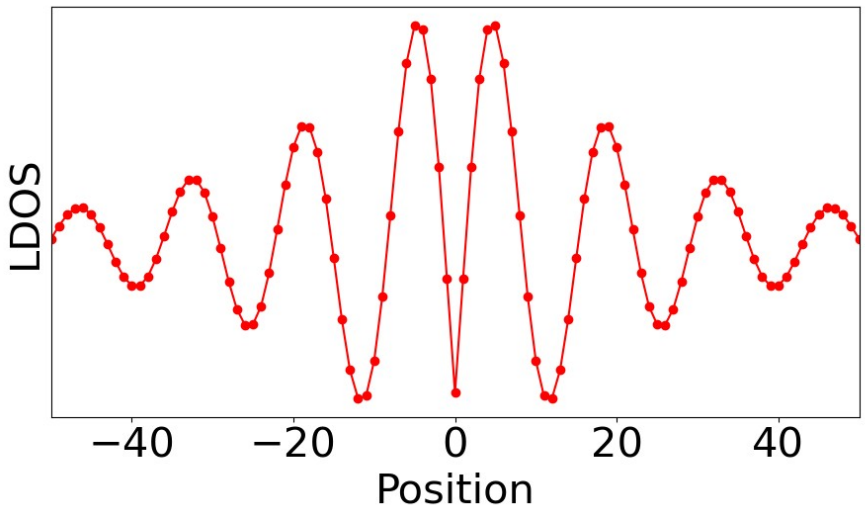
$$|k\rangle \sim \sum_x e^{ikx}|x\rangle$$

The change the density of states in real-space reflects the wavevector of the states at that energy

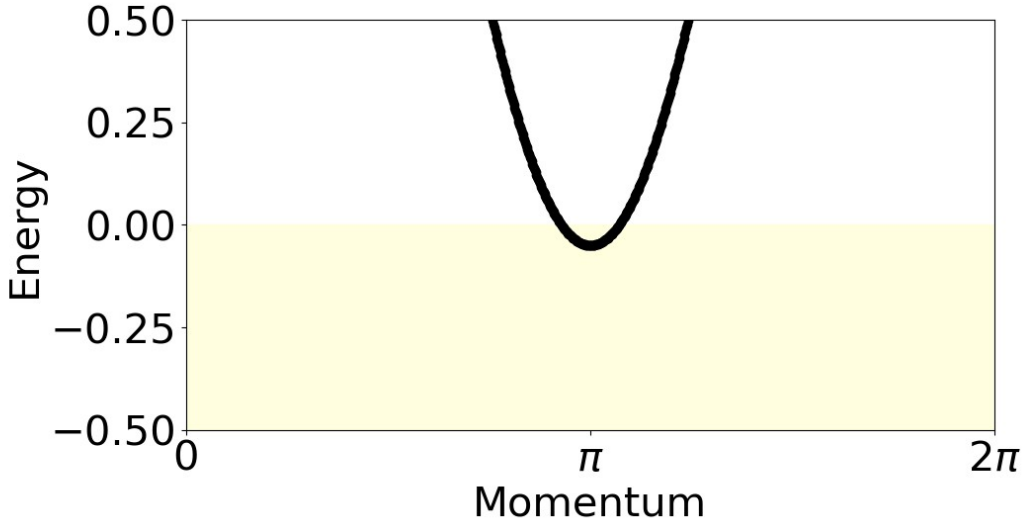
Impurity in a 1D metal

$$H = H_{metal} + V c_0^\dagger c_0$$

Local density of states



Original band structure

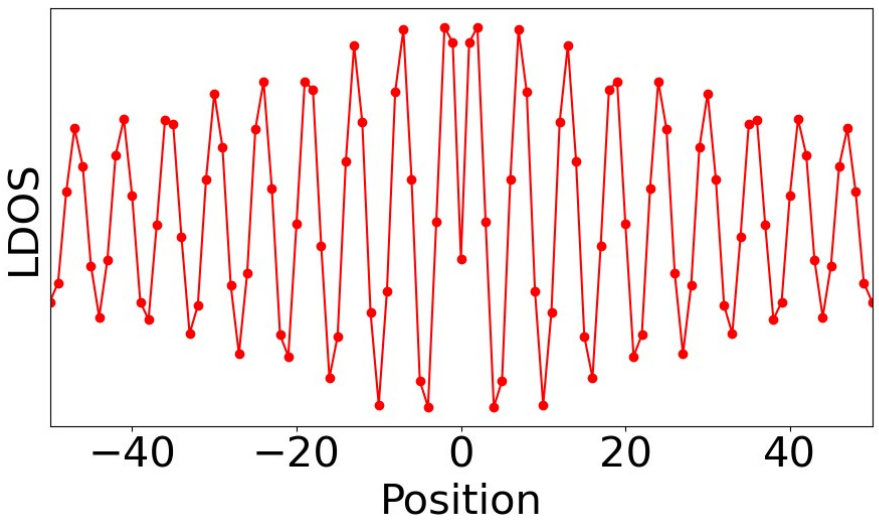


The density of states oscillated according to the Fermi wavelength

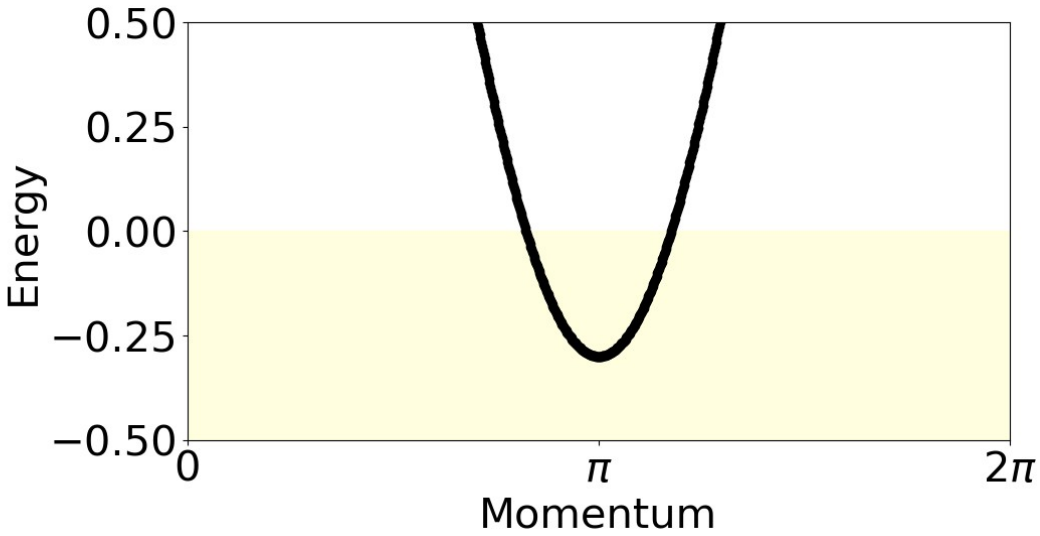
Impurity in a 1D metal

$$H = H_{metal} + V c_0^\dagger c_0$$

Local density of states



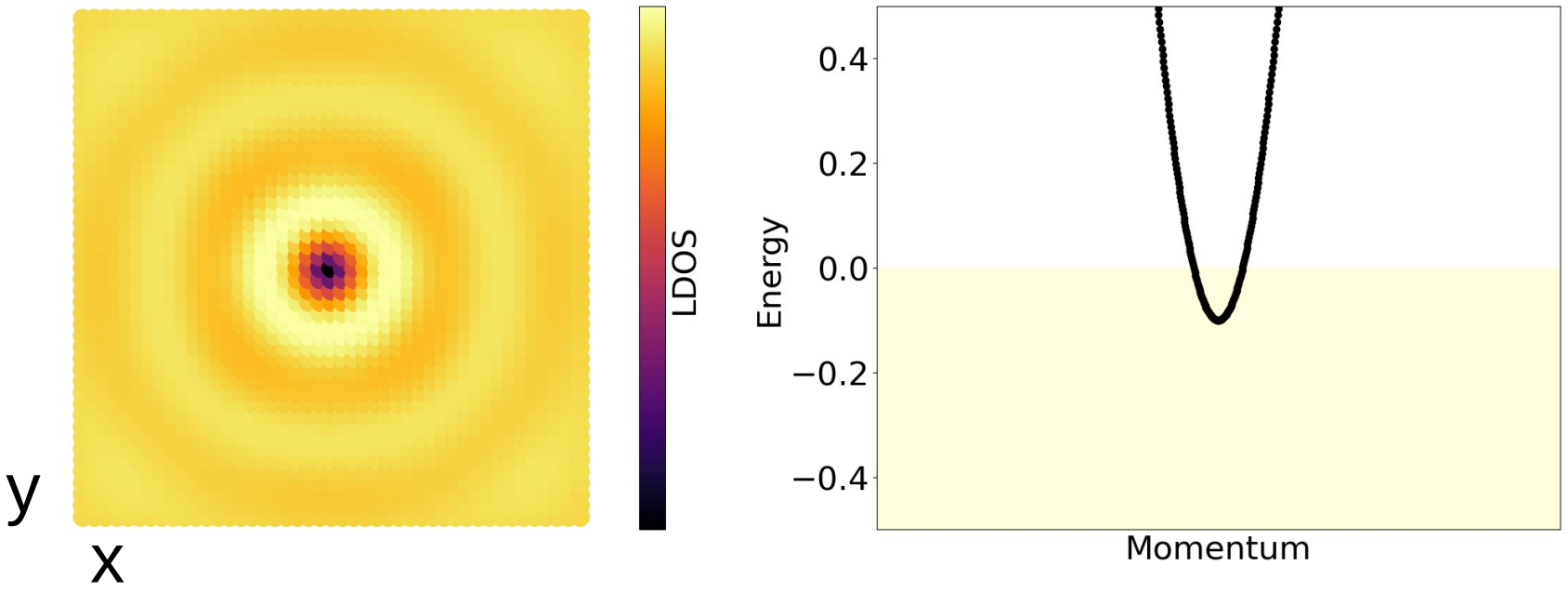
Original band structure



As the Fermi wavelength increases, the local DOS oscillates more quickly

Impurity in a 2D metal

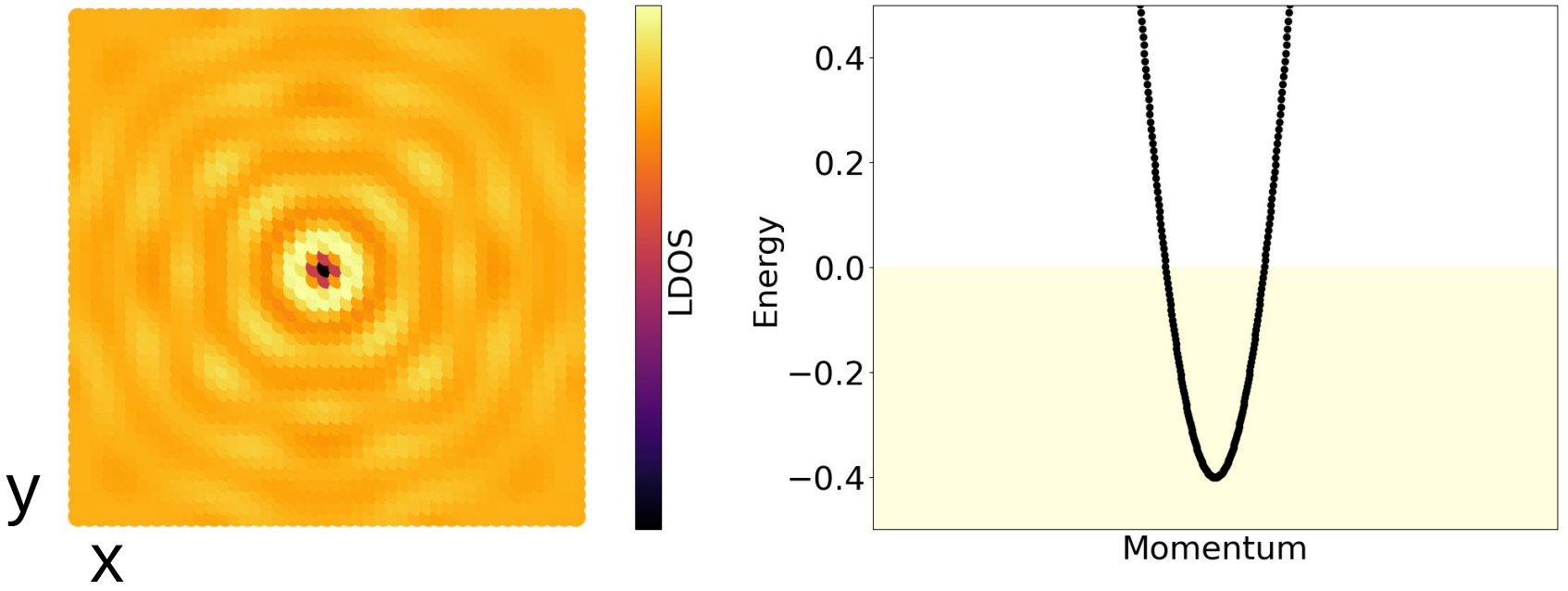
$$H = H_{metal} + V c_0^\dagger c_0$$



The oscillation of the local DOS is given by the Fermi wavevector

Impurity in a 2D metal

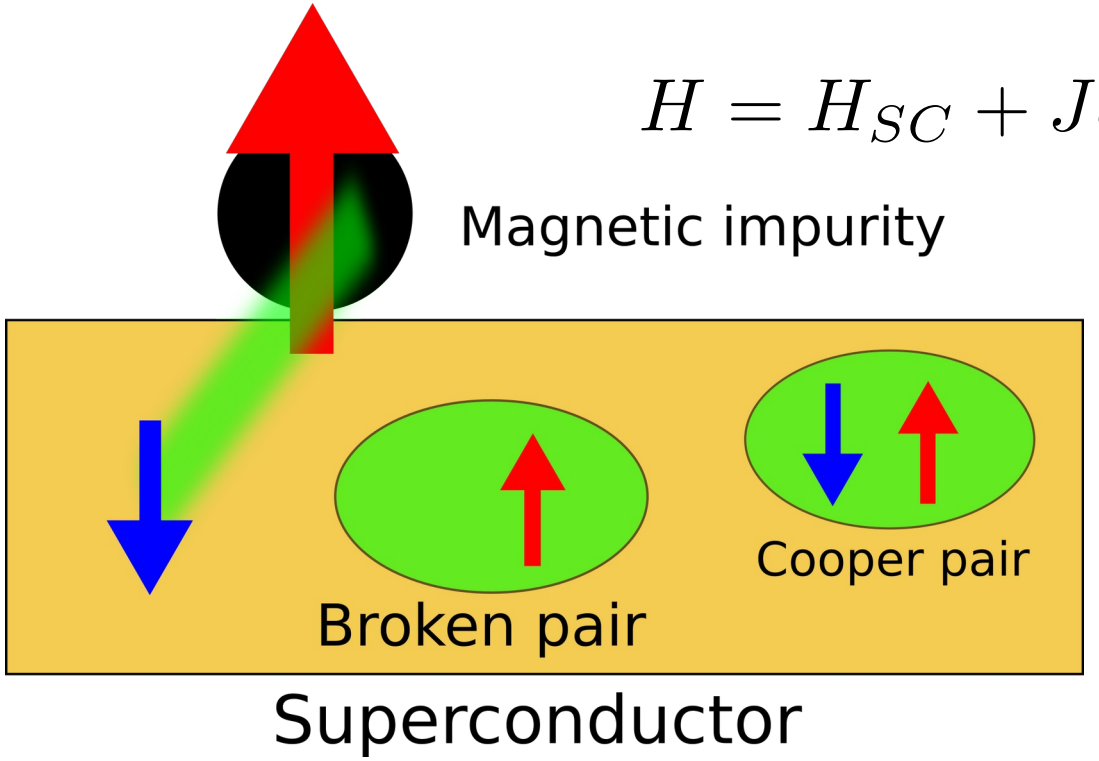
$$H = H_{metal} + V c_0^\dagger c_0$$



As the Fermi wavelength increases, the local DOS oscillates more quickly

Impurities in superconductors

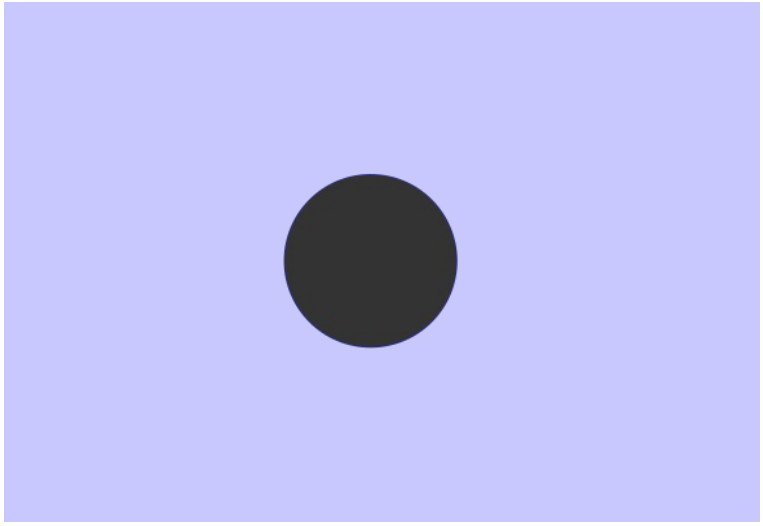
$$H = H_{SC} + J\sigma_z^{s,s'} c_{0,s}^{\dagger} c_{0,s'}$$



A magnetic impurity in a superconductor can break a Cooper pair, creating a bound state

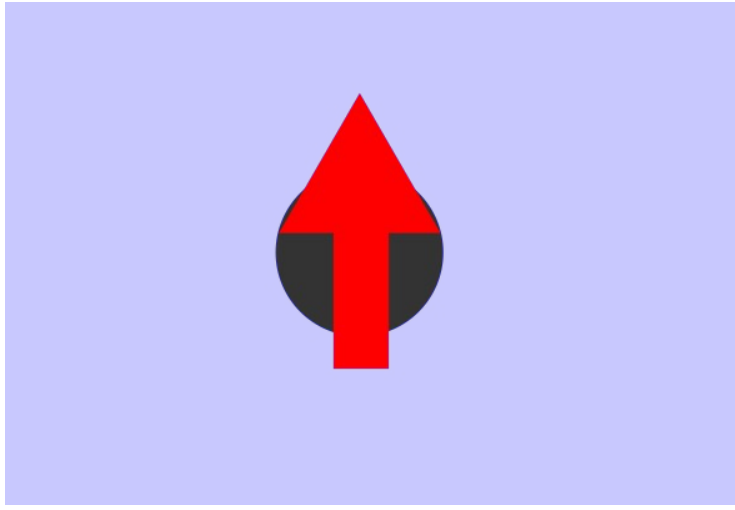
Two different types of impurities in conventional superconductors

A non-magnetic impurity



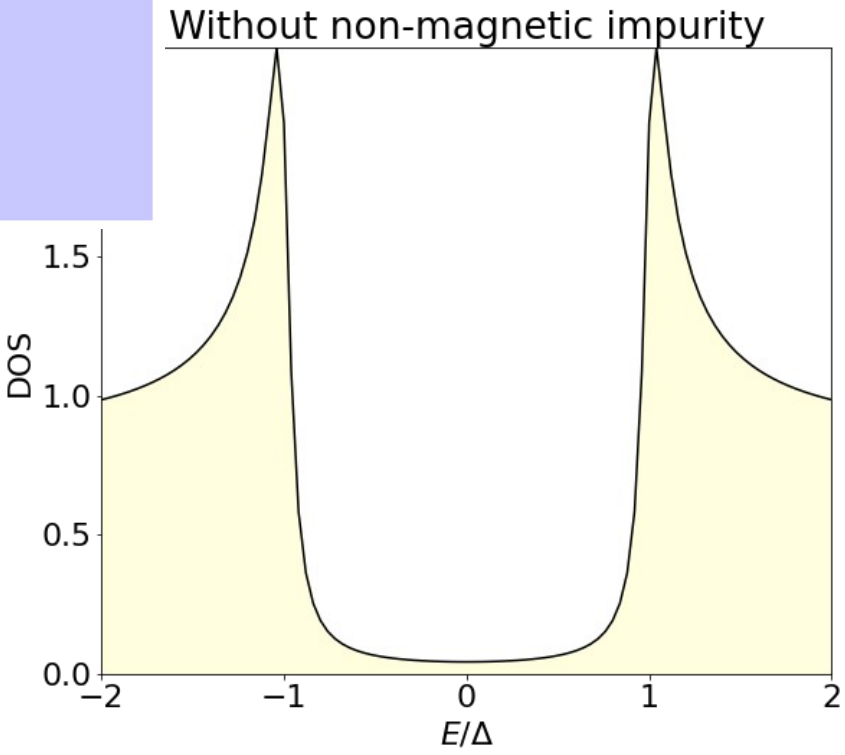
Does not break time reversal symmetry

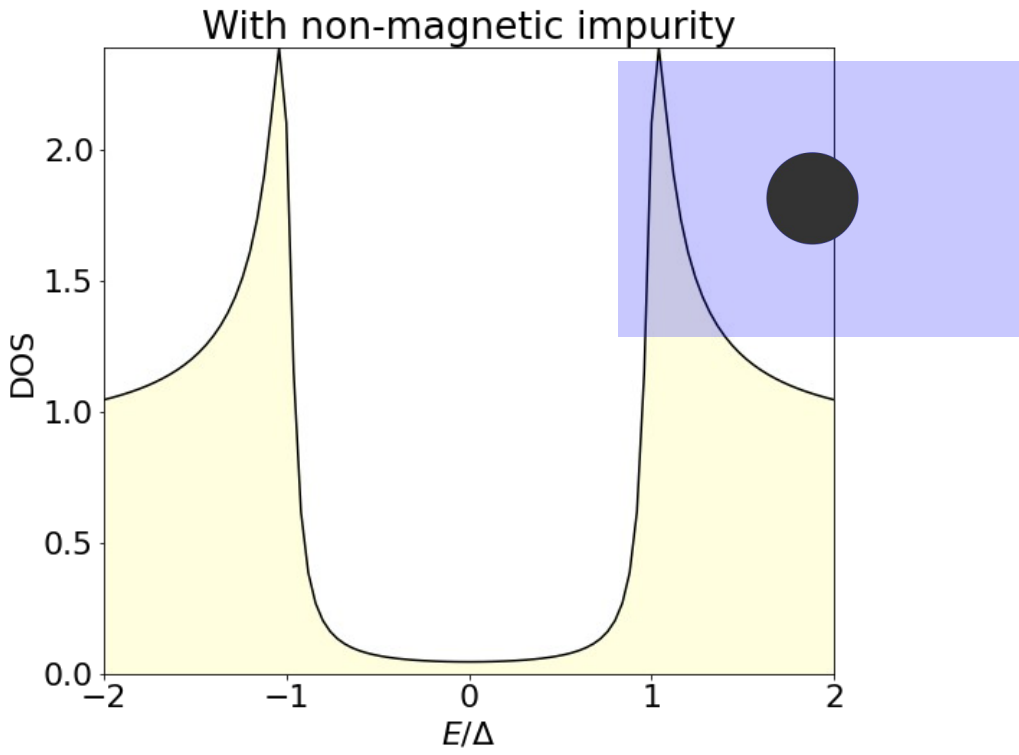
A magnetic impurity



Breaks time reversal symmetry

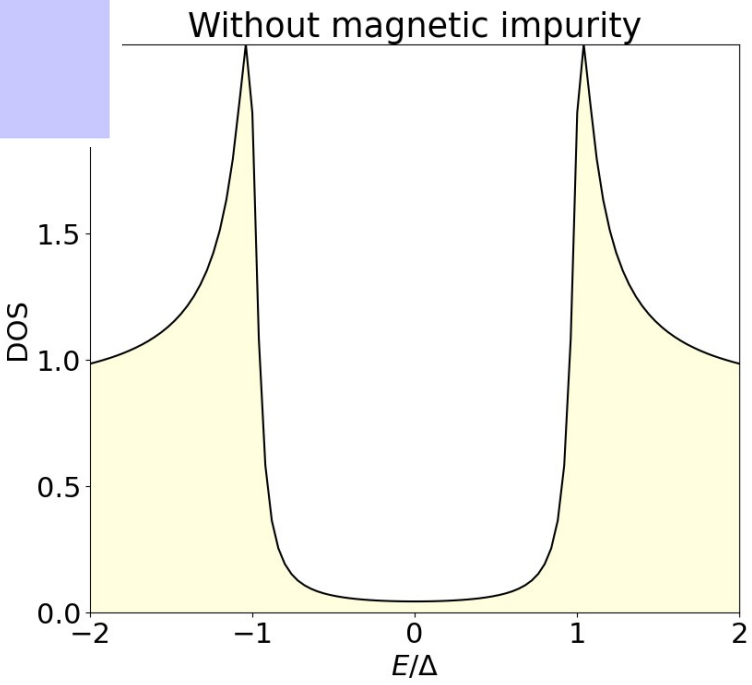
Non-magnetic impurity, conventional s-wave superconductor

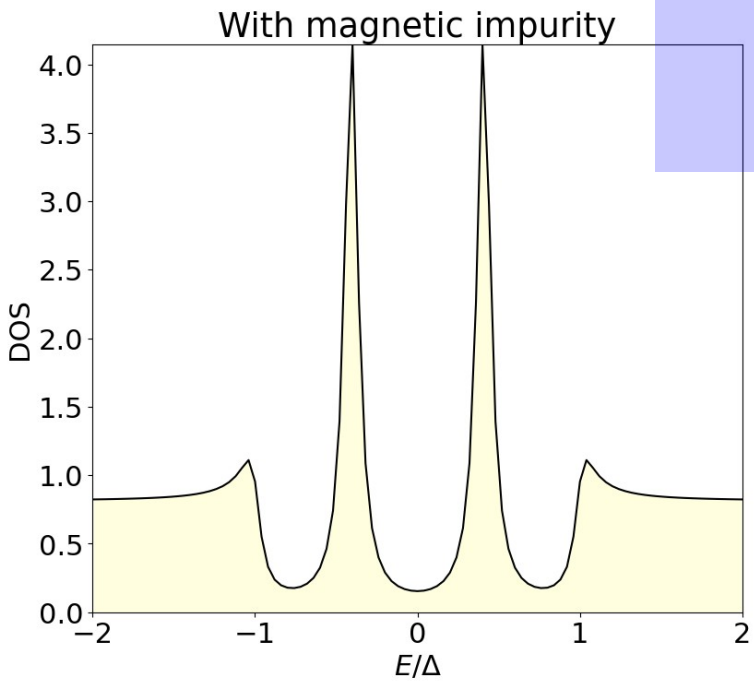




A non-magnetic impurity does not affect conventional s-wave superconductors

The interplay between magnetism and superconductivity



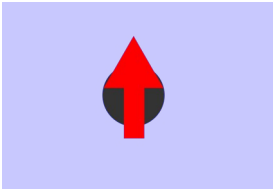


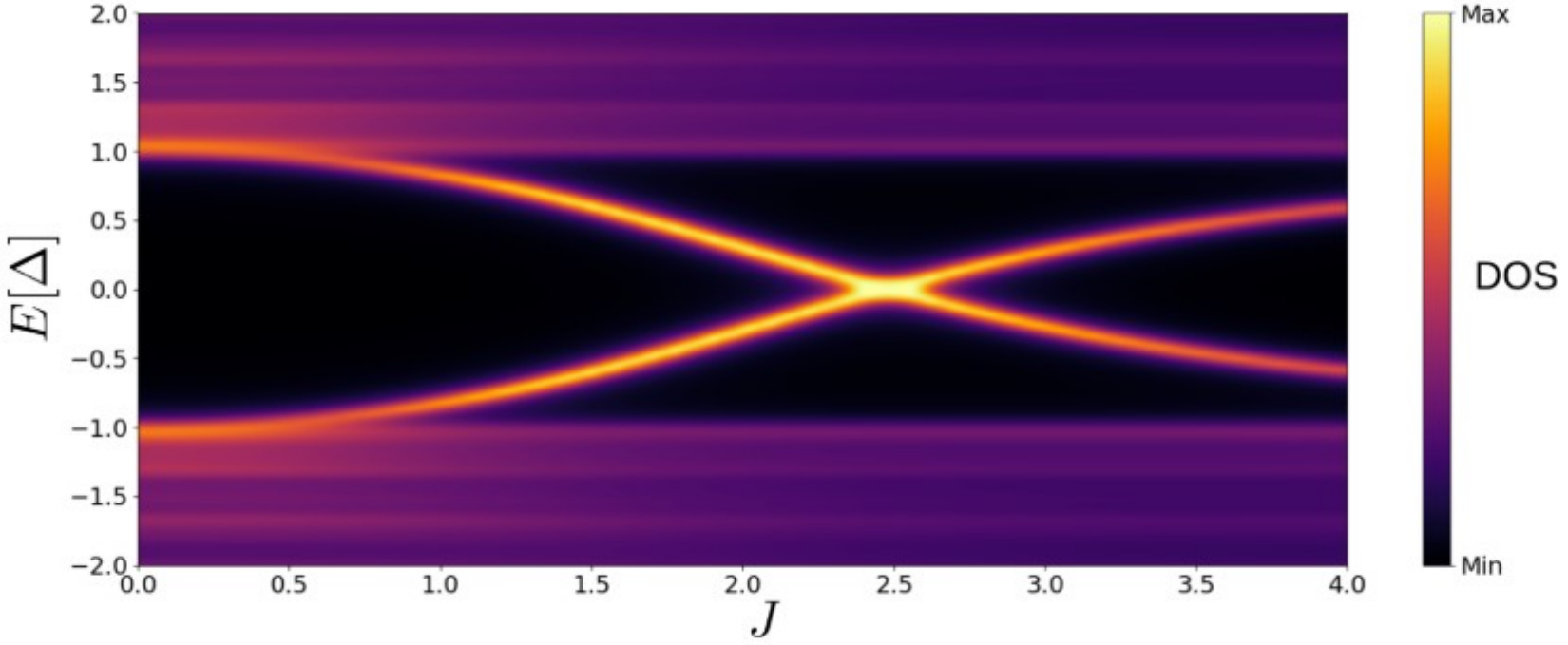


Magnetic impurities create in-gap states in fully gapped superconductors

Bound states by magnetic impurities in superconductors

The exchange coupling controls the energy of the in-gap state

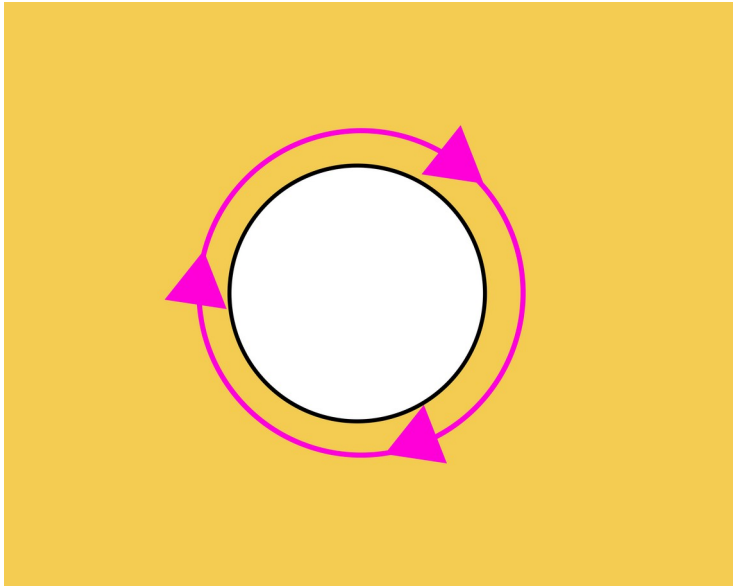




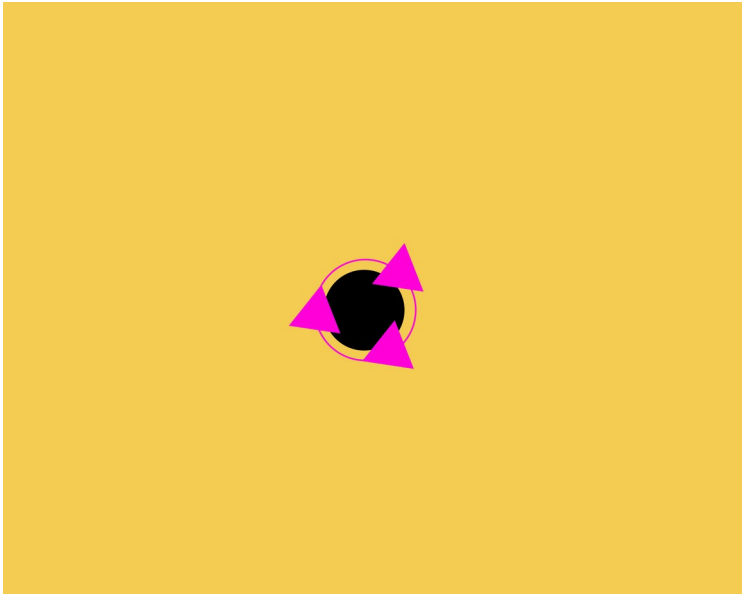
$$H = H_{SC} + J\sigma_z^{s,s'} c_{0,s}^{\dagger} c_{0,s'}$$

Impurities in topological matter

A physical hole create topological modes



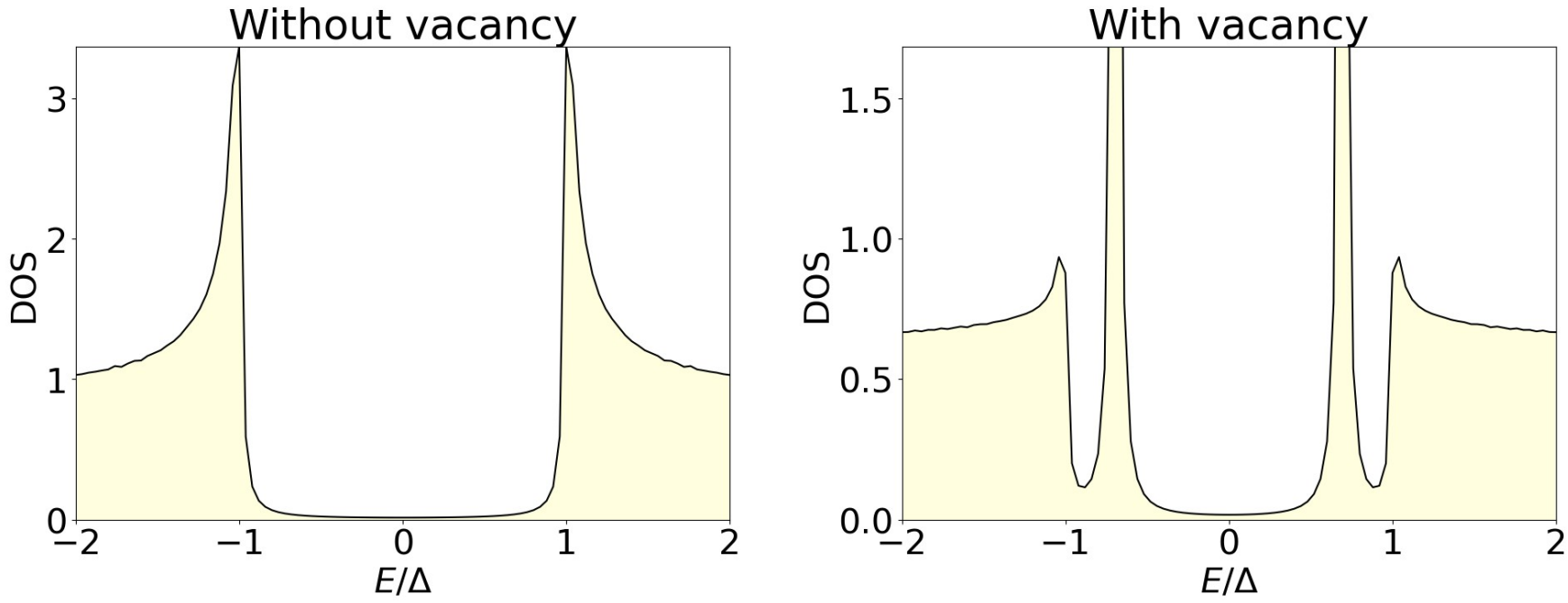
A vacancy can be understood as a very small hole



Topological matter develops in-gap states in the presence of vacancies

Vacancy in a topological superconductor

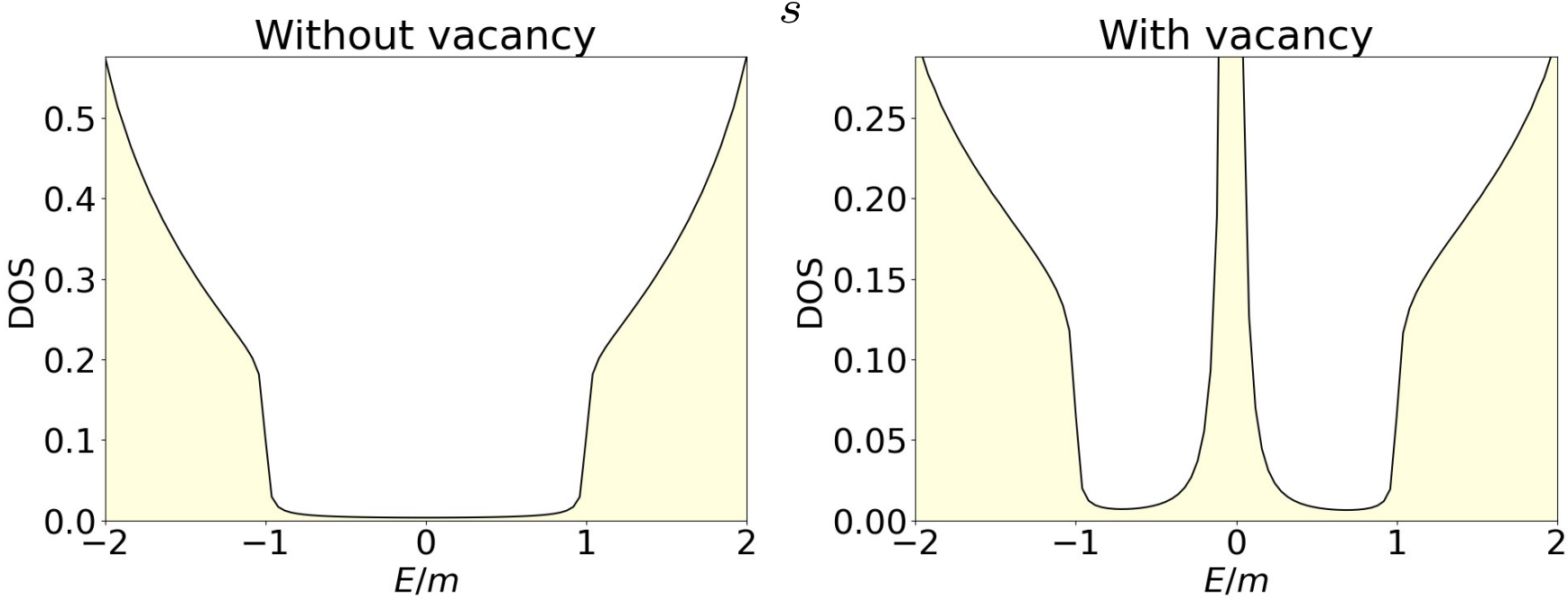
$$H = H_{SC} + V \sum_s c^\dagger_{0,s} c_{0,s} \qquad V \to \infty$$



In-gap modes appear when having a vacancy in a topological superconductor

Vacancy in a topological Chern insulator

$$H = H_{Chern} + V \sum_s c^\dagger_{0,s} c_{0,s} \qquad V \to \infty$$



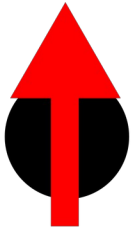
In-gap modes appear when having a vacancy in a topological insulator

Impurities in quantum magnets

Let us assume that a certain Hamiltonian realizes a QSL $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

A quantum magnetic state fulfills $\langle \vec{S}_i \rangle = 0$

What happens if we put a local ordered magnet in contact with a quantum magnet?



Magnetic impurity

Quantum magnet

Impurities in quantum magnets

We take the Hamiltonian of a quantum magnet with a local magnetic perturbation as

$$H = J \sum_n \vec{S}_n \cdot \vec{S}_{n+1} + J S_0^z$$

Pristine quantum magnet

Local magnetic moment

This a quantum many-body system, yet for S=1/2 we can approximate it with auxiliary fermions

$$S_i^\alpha = \frac{1}{2} \sigma_{s,s'}^\alpha f_{i,s}^\dagger f_{i,s'}$$

$$\sum_s f_{i,s}^\dagger f_{i,s} = 1$$

Impurities in quantum magnets

The screening of a S=1/2 1D quantum magnet

$$H = J \sum_n \vec{S}_n \cdot \vec{S}_{n+1} + J S_z$$

Is analogous to the screening of a 1D fermionic gas

$$H \sim \gamma \sum_n f^\dagger_{n,s} f_{n+1,s} + h.c. + J \sigma_z^{s,s'} f^\dagger_{0,s} f_{0,s'}$$

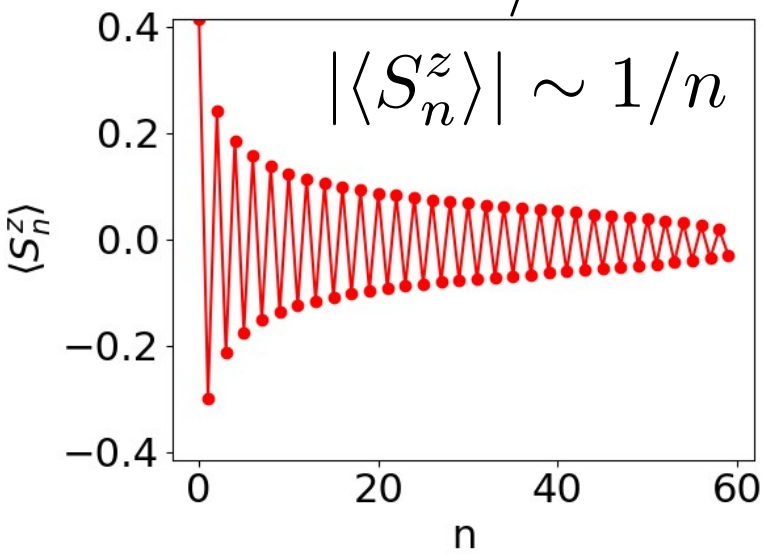
Is there a difference between S=1/2 and S=1 quantum magnets?

Magnetic screening in quantum magnets

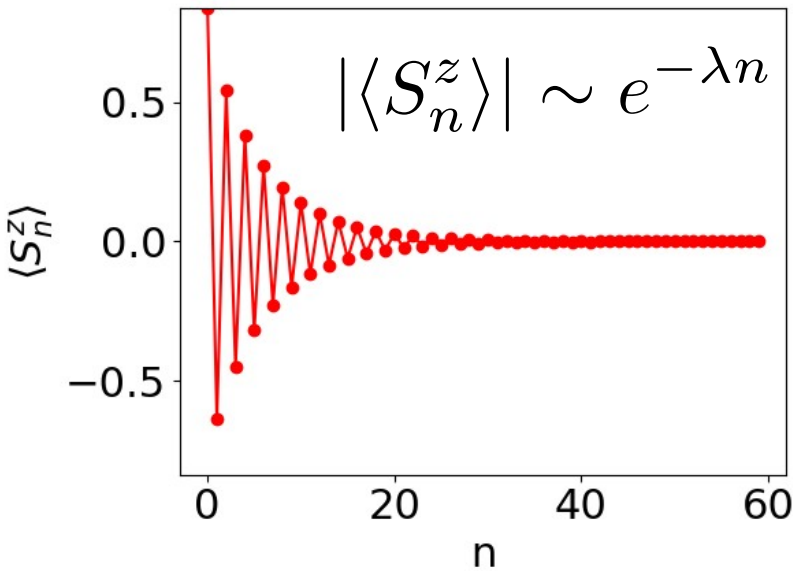
Let's address how a quantum magnet screens a magnetic moment

$$H = J \sum_n \vec{S}_n \cdot \vec{S}_{n+1} + J S_0^z$$

$S = 1/2$



$S = 1$

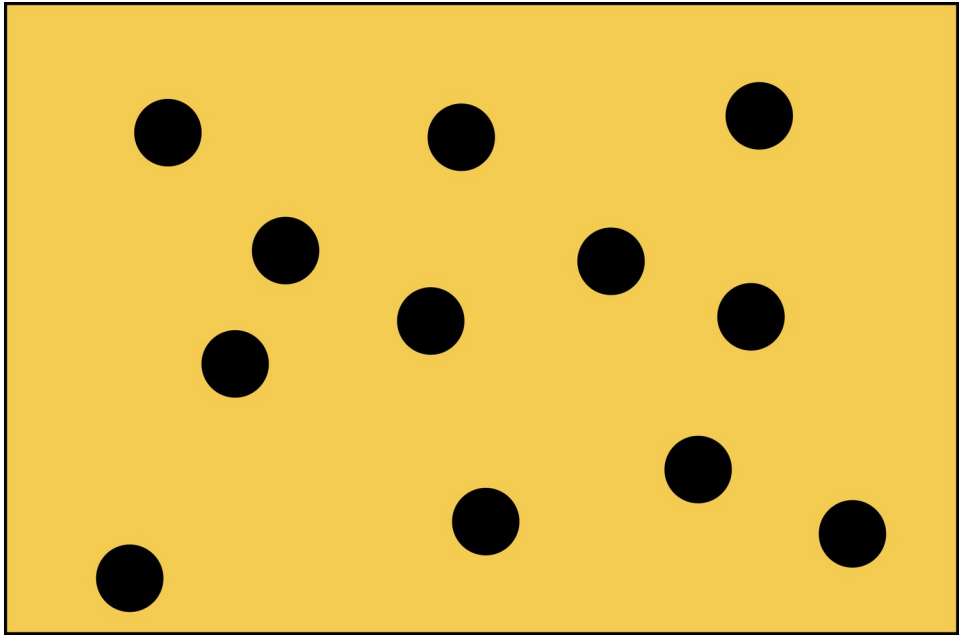


S=1 screens the impurity moment more effectively due its gapped nature

Disorder quantum materials and Anderson insulators

Breaking translational symmetry

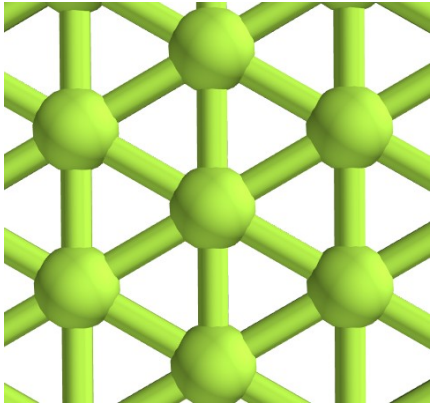
Impurities break the translational symmetry of the Hamiltonian



How are the ground state affected by the breaking of translational symmetry?

Three possible status of translational symmetry

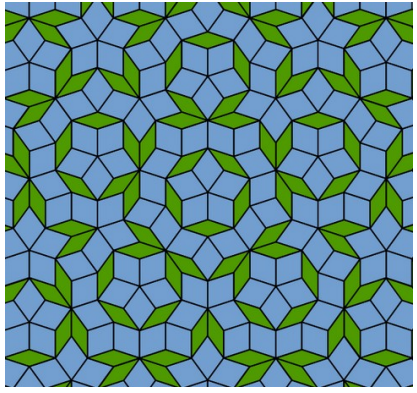
Periodic system



Wavefunctions
are delocalized

$$|\Psi(\mathbf{r})| = 1/\sqrt{N}$$

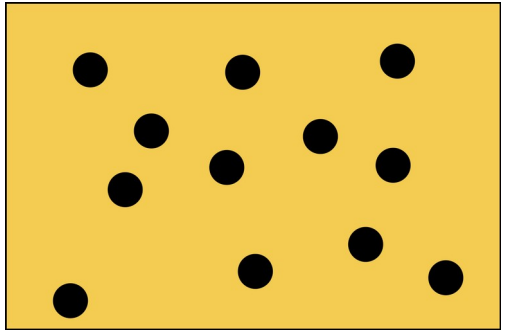
Quasiperiodic system



Wavefunctions
are delocalized,
critical or localized

$$|\Psi(\mathbf{r})| = |r|^{-\alpha}$$

Strongly disordered systems

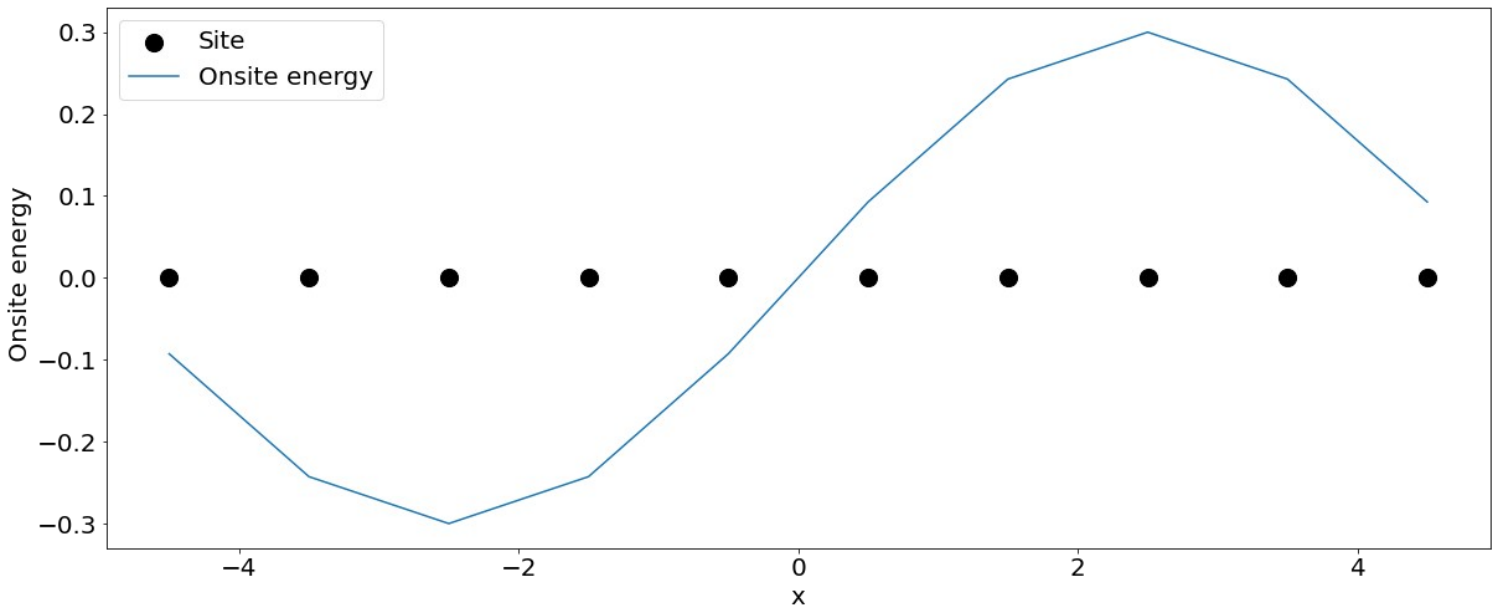


Wavefunctions
are localized

$$|\Psi(\mathbf{r})| = e^{-\lambda|r|}$$

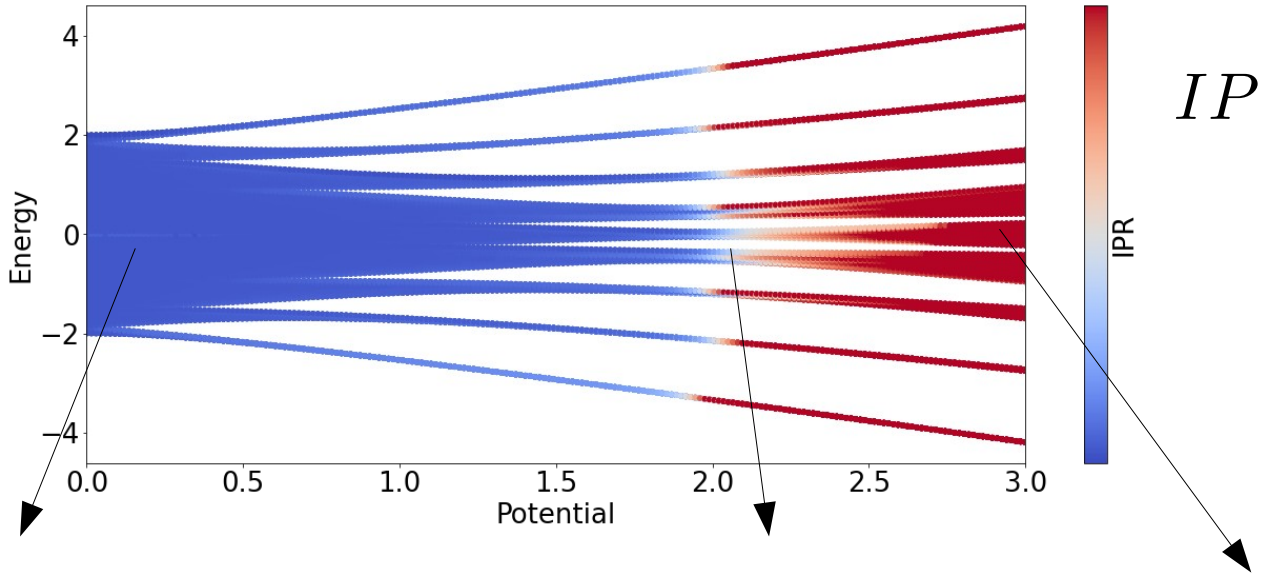
A quasiperiodic model

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + \lambda \sum_n \cos(qn) c_n^\dagger c_n$$



If q is an irrational number, the system is never periodic, but also not random

Quasiperiodicity, extended and localized states



$$IPR = \sum_r |\Psi(r)|^4$$

Extended states

$$|\Psi(\mathbf{r})| = 1/\sqrt{N}$$

Critical states

$$|\Psi(\mathbf{r})| = |r|^{-\alpha}$$

Localized states

$$|\Psi(\mathbf{r})| = e^{-\lambda|r|}$$

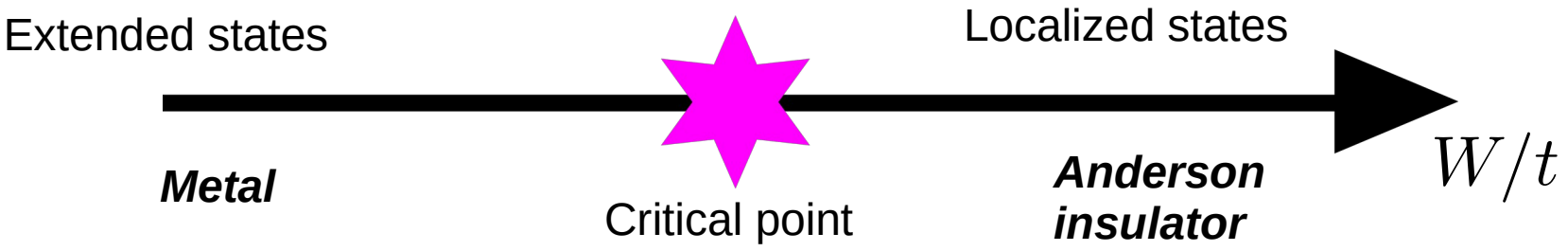
Disordered systems

In general, the Hamiltonian of a diosrdered system takes the form

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + W \sum_i r_i c_i^\dagger c_i \qquad r_i \in (-1, 1)$$

Periodic hopping

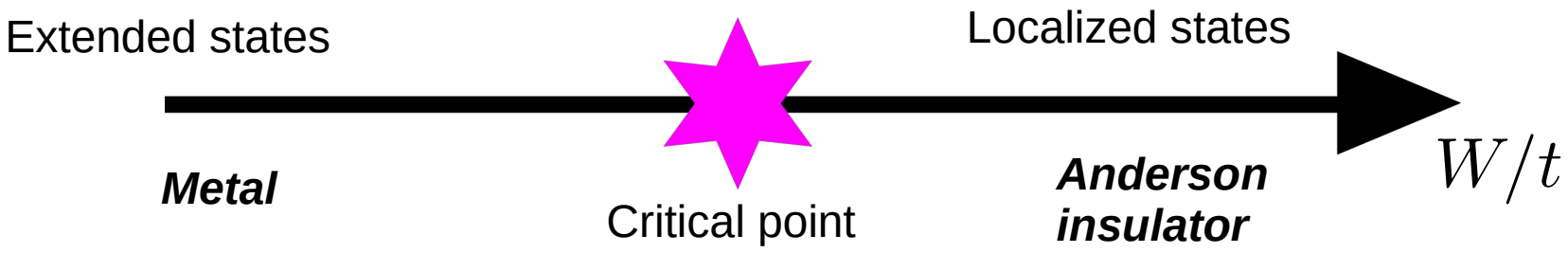
Random potential (disorder)



In general, a system becomes insulating at a critical value of the disorder

Disorder in 1D systems

In one dimensional systems, an arbitrarily small amount of disorder localizes the states



$$W_C/t \to 0$$

Take home

- The response to perturbations of a quantum material reflects the nature of its state
- The electronic structure determines the response to fluctuations
- Impurities in quantum materials allow to infer the nature of the underlying quantum state